Prabhavit Gupta

Senior ML Engineer: Agentic Systems, Rigorous Evals & Production AI Pipelines

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (Remote Europe / EU Work Authorization)

Summary

Machine Learning Engineer with 3 years of commercial production experience designing, deploying, and rigorously evaluating production AI and agentic LLM systems. Experienced in architecting end-to-end multi-step agent workflows with structured tool calling, deterministic validation guards, and offline/online evaluation harnesses (LLM-as-a-judge, golden benchmark datasets, regression testing). Specializes in taking unstructured domain data and clinical/document intelligence problems through exploration, serving, and continuous monitoring on cloud infrastructure. Dedicated to building reliable, auditable, and production-grade ML pipelines with Python, PostgreSQL, and modern containerized CI/CD.

Experience

FlowSpace.co • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and deployed agentic LLM workflows and Python inference pipelines handling document parsing, clinical/financial record extraction, and semantic categorization for 4,500+ active enterprise users.
- Designed and implemented multi-step agent orchestration with structured function calling, dynamic context retrieval (RAG), and deterministic fallback guards to prevent hallucination in critical data flows.
- Established automated evaluation harnesses using golden benchmark datasets and LLM-as-a-judge scoring protocols to detect accuracy drifts, schema regressions, and semantic edge cases prior to deployment.
- Built asynchronous task workers and queuing infrastructure with idempotency controls and exponential backoffs to guarantee fault tolerance and high system availability under peak loads.
- Integrated PostgreSQL relational schemas, embedding indexing, and vector similarity search for rapid retrieval across high-volume enterprise records.
- Maintained automated CI/CD deployment pipelines, unit/integration testing suites (pytest), and Docker containerized services on AWS and Linux.
- *Tools: Python, PostgreSQL, Hugging Face, PyTorch, OpenAI API, Anthropic, Docker, AWS, Supabase, Linux*

Urlink • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Built automated Python data pipelines and entity extraction routines aggregating disparate external API payloads into normalized PostgreSQL structures.
- Implemented automated validation rules, schema conformance checks, and continuous regression testing across customer data feeds.
- Partnered closely with cross-functional product stakeholders to define interface specifications, SLA metrics, and operational monitoring dashboards.
- *Tools: Python, REST APIs, PostgreSQL, Supabase, TypeScript, Linux*

Independent Technical Consultant • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Developed custom Python automation services and webhook processors for automated unstructured text extraction, data classification, and downstream API syncing.
- Built offline test harnesses and synthetic test-case generators to benchmark model output stability across prompt iterations.
- Formulated operational runbooks, failure mode analyses, and comprehensive technical documentation for non-technical stakeholders.
- *Tools: Python, Webhooks, REST APIs, Supabase, Linux*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Integrated machine learning models and automated email notification webhooks, establishing end-to-end data validation across user registration and onboarding journeys.
- *Tools: Python, OpenAI API, Sendgrid, REST APIs*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** European Long Term Residence Permit (Italy), unrestricted EU work authorization (Partita IVA / Italian employment ready). No sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (beginner).
- **Availability:** Available immediately for full-time remote roles in Europe.

Core Skills

- **Machine Learning & Agent Systems:** Multi-step agent orchestration, tool use, prompt engineering, RAG, embeddings, vector similarity search, Hugging Face, PyTorch, OpenAI/Anthropic APIs.
- **Evaluation & Quality Engineering:** Offline eval harnesses, LLM-as-a-judge pipelines, golden test datasets, regression testing, drift detection, structured schema validation.
- **Backend & Data Systems:** Python 3, asynchronous processing, PostgreSQL schema design, indexing, SQL optimization, Redis, REST APIs, JSON schema validation.
- **Infrastructure & Reliability:** Docker containerization, AWS, Linux system administration, GitHub Actions CI/CD, pytest, structured logging and monitoring.
- **Engineering Practices:** End-to-end project ownership, clean architecture, ambiguity resolution, cross-functional technical communication.